School of Computing

CT2- 21CSS201T -Computer Organization and Architecture

Part -A MCQ (10*1 = 10 Marks) Total : 50 marks

1. Assume the main memory size is 64 kb, compute the number of address lines needed to address the main memory.

   a. 8 bit
   b. 16 bit
   c. 24 bit
   d. 32 bit

   Ans: 16bit

2. Identify the addressing mode of the instruction
   JC Label

   a. Direct addressing
   b. Index addressing
   c. Relative addressing
   d. Immediate addressing

   Ans: Relative Addressing

3. In which type of addressing there is no memory related operation?

   a. Register indirect addressing

   b. Register addressing

   c. Direct addressing

   d. Indirect addressing

   Ans: Register addressing

4. How many full adders are required to design an N-bit binary adder?

   a. n/2
   b. n
   c. 2 x n
   d. n^2

   Ans: n

5. _____ reduces the number of summands by a factor of 2.

a. Carry save addition
b. Bit pair recoding
c. Multiple pair recoding
d. Carry look ahead circuit

Ans: Bit pair recoding.

6. What is the relation between sign exponent E and bias exponent E' in single precision floating point number?

a. E=E'-1023
b. E=E'+127
c. E=127-E'
d. E=E'-127

Answer: E= E'-127

7. The multiplier 11010 is recoded as _____

a. 0 -1 -1 0 0
b. 0 -1 +1 +1 0
c. 0 -1 +1 -1 0
d. 0 +1 -1 -1 0

Answer: 0 -1 +1 -1 0

8. The carry equation for full adder is _____

a. XY′+X′Z+Y′Z
b. X'Y+XZ'+YZ
c. XY'+XZ+YZ'
d. XY+XZ+YZ

Answer: d. XY+XZ+YZ

9. How many select lines are required for 64 : 1 Multiplexer

a. 8 lines
b. 6 lines
c. 4 lines
d. 16 lines

Ans: b. 6 lines

10. In Booth's recoding, M is multiplicand and -1 is booth recoded multiplier, then what will be the result of multiplication?

a. 1′s complement of M
b. 2′s complement of M
c. M
d. Right shift of M

Ans: b. 2′s complement of M

Part –B (Answer any four) (4 x 4 Marks = 16 marks)

11. Explain the following instructions (each 2marks)

a. ADD R3, - (R4)
b. ADD R1, (R2)+

a. R4 is **decremented before accessing the operand**, then it is read, added with R3 and result stored in R3.

b. The content pointed by the address stored in R2 is added with R1 and the result is stored in R1. **After the operation the value of R2 is incremented**.

Auto decrement mode: Effective address of the operand is the contents of a register specified in the instruction. Before accessing the operand, the contents of this register are automatically decremented to point to the previous consecutive memory location.

Auto increment mode: Effective address of the operand is the contents of a register specified in the instruction. After accessing the operand, the contents of this register are automatically incremented to point to the next consecutive memory location

12. What is the total memory capacity 8086 processor can address. Write a short note of Memory segmentation used in 8086.

Ans: Total memory capacity is 2^{20} = 1MB. **(1 mark)**

Memory is divided into 4 segments. Size of each segment is 256KB **(3 marks)**

Code Segment - Contains the instruction code of a program.
Data Segment - Stores data, variable and constants
Stack Segment - Stores interrupt and subroutine return addresses.

Extra Segment. - Data related to string operations.

13. In a 4 bit Carry look ahead adder, if the input bits are 1011 and 1001 what is the generate, propagate and carry signal from the third stage. [4 Marks]

Ans:

$G_i = A_i B_i$

$P_i = A_{i\,(XOR)} B_i$

$$C_o = G_o + P_o C_{in}$$
$$= A_o B_o + (A_o \oplus B_o)\, C_{in}$$
$$= 1.1 + (1 \oplus 1)0 \;\; [as\; C_{in} = 0\,]$$
$$= 1$$

$$C_1 = G_1 + P_1 C_0$$
$$= A_1 B_1 + (A_1 \oplus B_1)\, C_0$$
$$= 1.0 + (1 \oplus 0)1$$
$$= 1$$

$$C_2 = G_2 + P_2 C_1$$
$$= A_2 B_2 + (A_2 \oplus B_2)\, C_1$$
$$= 0.0 + (0 \oplus 0)1$$
$$= 0$$

generation of 3rd stage = $G_2 = A_2 . B_2 = 0.0 = 0$
Propagation of 3rd stage = $P_2 = A_2 \oplus B_2 = 0 \oplus 0 = 0$

so the generate, propagate and carry signal from the third stage are 0,0,0 respectively

14. Represent 350.2532 in single precision format.

350.2532 = 101011110.010000001101000 (2 marks)

Normalize: 1.01011110 010000001101000 x 2^8

Ans: **0 10000111 01011110 010000001101000** (2 marks)

15. Using Booth's recoding technique multiply +10 and +12.

Multiplicand = + 10
Multiplier = + 12

Binary value of 10 = 01010

Binary value of 12 = 01100

Recoded value of 12 = +1 0 -1 0 0

10 - 0 1 0 1 0

12 - +1 0 -1 0 0

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0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 x
1 1 1 1 0 1 1 0 x x
0 0 0 0 0 0 0 x x x
0 0 1 0 1 0 x x x x

---

1 0 0 0 1 1 1 1 0 0 0

Decimal equivalent = 120

Part - C (12 x 2 = 24 marks)

16. Define addressing modes. With examples explain the following modes. (2+10)

(i) Register addressing mode
(ii) Immediate
(iii) Indirect addressing
(iv) Relative addressing
(v) Index addressing.

1. **Register addressing mode-**

In this mode, operands are specified using registers. This addressing mode is normally preferred because the instructions are compact and fastest executing of all instruction forms.

Registers may be used as source operands, destination operands or both.

Example:

MOV AX, BX

This instruction copies the contents of BX register into AX register. AX ← BX

2. **Immediate addressing mode-**

In this mode, the operand is specified in the instruction itself. Instructions are longer but the operands are easily identified.

Example:

MOV CL, 12H

This instruction moves 12 immediately into CL register. CL ← 12H

3. **Indirect addressing mode:**

In this mode the address of the operand is stored indirectly in a Register.

Example: MOV CX, [BX]

4. **Relative addressing mode-**

In this the effective address is calculated with respect to Instruction Pointer.

Example: JNZ 8 bit address.

5. **Base indexed addressing mode-**

Here, operand address is calculated as base register plus an index register.

Example:

MOV CL, [BX + SI]

This instruction moves a byte from the address pointed by BX + SI in data segment to CL.

OR

B. Write a sequence of instructions that will compute the value of $y = x^2 + 2x + 3$ for a given x using (each 4 marks)

a. three-address instructions
b. two-address instructions
c. one-address instructions

Ans:

Three address instruction

```
Mul z, x, x
Mul y, 2, x
Add y, y, z
Add y, y, 3
```

Two address instruction

```
Mov z, x
Mul z, x
Mov y, 3
Add y, z
Mov z, x
Mul z, 2
Add y, z
```

One address instruction

```
Mov x
Mul x
Store z
Mov x
Mul 2
Add z
Add 3
Store y
```

17.a.Two person Ram and Krishna were playing a multiplier game with two numbers based on some condition, if least significant bit is 1, Krishna gets a chance to perform addition and shift, if least significant bit is zero Ram gets a chance to perform shift operation. List the number of chances received by Ram and Krishna. The numbers used by Ram and Krishna for the game was 13 and 12. (8 Marks)

Multiplicand (M) = 13 = 1101

Multiplier (Q) = 12 = 1100

C	A	Q	Action
0	0000	1100	
0	0000	0110	Shift A_Q right
0	0000	0011	Shift A_Q right
0 0	1101 0110	0011 1001	A ← A+M Shift A_Q right
1 0	0011 1001	1001 1100	A ← A+M Shift A_Q right

Answer = Contents of A_Q

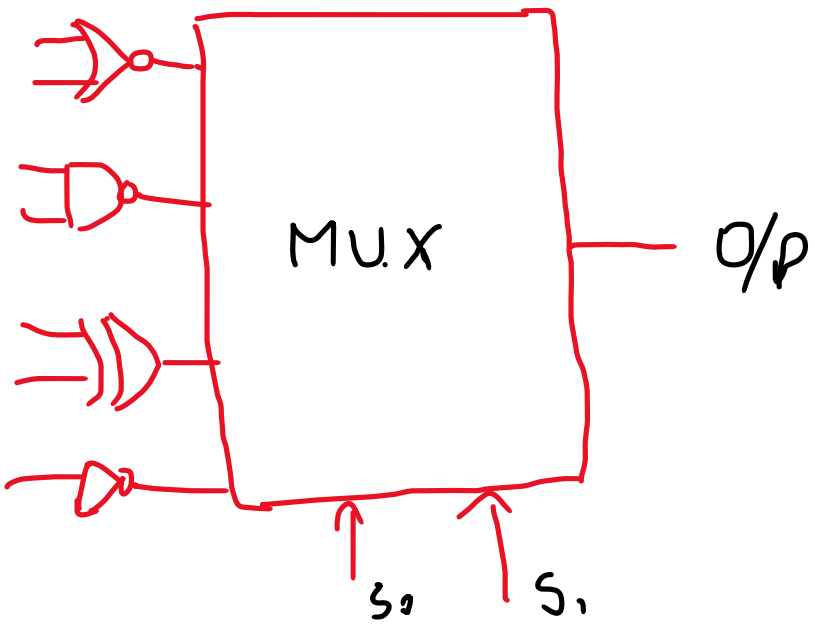
Answer: 10011100 = 156

Krishna gets two chances of addition and shifting

Ram gets two chances of shifting

17(ii). Design a 4-bit ALU with the following gates NOR, NAND, XOR and NOT gate.

(4 marks)



S_0	S_1	
0	0	Nor
0	1	Nand
1	0	exclusive OR
1	1	Not

(or)

B. In one rack of library 5 books can be placed, how many racks are needed for 12 books.? Perform the required operation based on the following given algorithm.: Do the following for n times: If the sign of A is 0, shift A and Q left one bit position and subtract M from A, otherwise, shift A and Q left and add M to A. Now, if the sign of A is 0, set q0 to 1, otherwise, set q0 to 0. Step 2: If the sign of A is1, add M to A.

Answer Key:

Non restoring Division

Remainder: 0010 in A register

Quotient: 0010 in Q register

M- Divisor – 00101

Q – Dividend - 1100

A	Q	Action
00000	1100	Initialize
00001	110_	Shift left A, Q
11100	110_	A← A-M
11100	1000	Q_0 ->0
11001	000_	Shift left A, Q
11110	000_	A← A+M
11110	0000	Q_0 ->0
11100	000_	Shift left A, Q
00001	000_	A← A+M
00001	0001	Q_0 -> 1
00010	001_	Shift left A, Q
11101	001_	A← A+M
11101	0010	Q_0 ->0

Since A is negative perform A← A+M 11101+00101 = 00010 (Discard carry)